

Chief Scientist Office 2026 Team Description Paper

Takaaki Numai, Airi Yokochi, Joshua Supratman, Tatsuro Sakaguchi, Yushi Kaida, Gakuto Okamoto, and Koshi Shibata*

SoftBank Corp. Chief Scientist Office
<https://www.robocup.ros-sier.com/>, <https://github.com/sbgisen>

Abstract This paper presents SoftBank Corp. Japan team Chief Scientist Office’s mobile manipulator “SOAR” for RoboCup 2026 @Home League. Designed as a research and development platform, SOAR advances robotic intelligence through modular hardware, integrated perception–planning–action architecture, and large language model (LLM)-driven behavior generation. The system combines robust 3D scene fitting, object recognition, and natural language interaction to enable adaptive task execution in domestic environments. Built on ROS 2, SOAR emphasizes reusability, openness, and long-term maintainability. This paper details the platform’s hardware modules, key software systems, and contributions to the robotics community, positioning SOAR as a foundation for future intelligent service robots.

1 Introduction

This paper details SoftBank Corp. Chief Scientist Office and our mobile manipulator “SOAR” for RoboCup @Home League 2026. This is our third appearance at RoboCup @Home having obtained 4th place in RoboCup 2024 @Home Open Platform League previously. Since our last participation in 2024, we have made significant improvements to our hardware and software system for mobile manipulator “SOAR”. Building upon previous iterations, we enhanced system integration by migrating to ROS 2 and introducing a robust perception-to-action pipeline encompassing 3D scene fitting, object recognition, speech interaction, and LLM-driven behavior planning. We also report our increasing contributions to the robotics community through open-source releases and conference sponsorships, and discuss the broader application of SOAR as a customizable and affordable robot platform for research and commercial use.

The team has developed various robots over the years with SOAR being one of them. SOAR serves as an internal research platform dedicated to advancing robotic intelligence, system integration and modular design. Its purpose is to accumulate technical knowledge, test new software architectures, and develop a foundation for the future robotic systems within SoftBank and beyond. The project’s long-term vision is to develop a customizable and affordable robot

* The order of authors is based on age.

platform - one that democratizes robotics by making it accessible, adaptable, and sustainable for both research and commercial applications. SOAR functions as a stepping stone toward this vision, enabling iterative exploration of technologies that lower the barrier to robotic deployment in diverse domains.

1.1 Focus of research

Our research centers on unifying perception, planning, and interaction under a modular and extensible architecture. Our current research focuses on the following areas:

- **Modularity and Configurability:** Investigate how to design a system that allows for easy integration and customization of hardware modules.
- **Device Integration:** Tackling the hardware challenges related to the seamless integration of diverse devices and components
- **Integrated Perception–Planning–Action Framework:** Building a cohesive software architecture where perception results directly inform task planning and control, allowing the robot to react contextually to real-world changes.
- **LLM-Driven Behavioral Planning:** Exploring large language models as high-level planners capable of transforming natural language goals into structured robot behaviors, enabling flexible and adaptive responses beyond pre-defined rules.

Through these research directions, SOAR contributes to SoftBank’s broader mission: building a practical, affordable, and modular robotics platform that accelerates innovation, fosters collaboration, and democratizes access to advanced robotic technologies.

The following sections detail how these research themes are implemented through SOAR’s modular hardware and integrated software.

2 Hardware Modules

SOAR’s hardware architecture emphasizes flexibility, maintainability, and re-configurability, allowing researchers to swap components or integrate new sensors and actuators with minimal effort. The mobile manipulator is comprised of four main components: the mobile base, the torso, the manipulator, and the head as shown in Fig. 1. The manipulator is a commercially available product, while the mobile base, the torso, and the head are custom-developed by our team. The head and the torso are separate components designed for specific functionalities; we designed the head to accommodate various interchangeable sensors and the torso to enhance the manipulator’s reach and versatility. All components are designed with minimal cable protrusion, with circuits centralized on the torso, thus streamlining the connection across components with minimal cabling. This modular approach not only facilitates customization but also organizes and simplifies maintenance and upgrades, allowing for the interchange of components as needed.

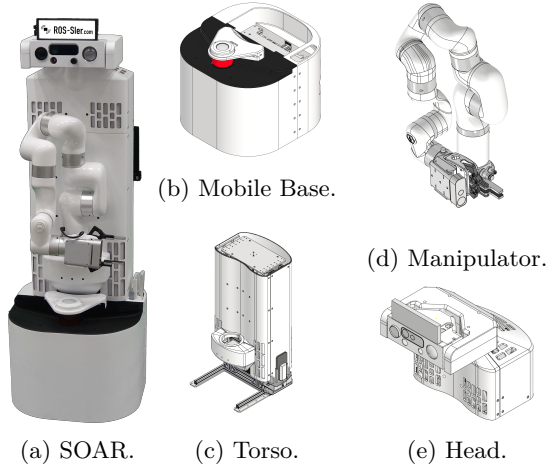


Figure 1: SOAR and its four components.

2.1 Mobile Base

SOAR’s mobile base (Fig. 1b) is a compact 2-wheel differential drive platform, designed for indoor navigation. Both driving wheels and casters are equipped with suspension systems to ensure smooth movement and stability. The mobile base integrates 2D LiDAR sensors and a 9-axis IMU for localization and autonomous navigation. It also includes a battery management system and a docking charging connector, enabling the SOAR to recharge itself without human intervention thus extending its operational time. The motor drivers and the battery management system are connected to a USB hub, resulting in only two cables extending out: the USB cable from the device hub and the power supply cable.

Previously, SOAR used a commercially available mobile base with custom modifications. For RoboCup @Home 2026, we swapped with a newly developed, in-house base that has smaller footprint and equipped with a built-in LED button for direct user interface.

2.2 Torso

The torso (Fig. 1c), developed by our team, is a one degree of freedom linear actuator designed to vertically adjust the position of the head and manipulator. It incorporates a constant-force spring to provide gravity compensation, reduces motor load and improves energy efficiency during vertical movement. This mechanism provides an additional 350 mm of vertical clearance, significantly enhancing the manipulator’s reach and versatility. The torso also serves as the central hub for SOAR’s hardware components, housing the main PC and controller box.

2.3 Manipulator

The manipulator (Fig. 1d) is xArm 7¹, a commercially available product developed by uFactory. The end-effector is equipped with a two-finger gripper and a RGB-D hand camera.

2.4 Head

The head (Fig. 1e), designed as SOAR’s primary sensory and interactive module, serves as a centralized platform for perception and user engagement. The module supports various range of sensors and user interface devices for environmental understanding and social interaction.

Equipped with a USB hub for efficient cable management and a separate USB port for the high-bandwidth depth camera, it allows for easy swapping of sensors and devices, ensuring that the component can be customized for specific requirements. The head extends three specific cables: a USB cable from the sensor hub, a separate USB for the depth camera, and a power supply cable.

In addition, the head has 2 degrees of freedom for pan and tilt movements, allowing SOAR to dynamically adjust its field of view for environment scanning, object tracking, and social interaction.

3 Software

SOAR’s software architecture is built on ROS2 Jazzy, leveraging its modularity and extensive ecosystem to facilitate development and integration of various functionalities as well as long-term maintainability comparing to ROS1. Here we introduce several key software modules that form the core of SOAR’s perception, interaction, and behavior systems, which we use in RoboCup @Home tasks.

3.1 Object Recognition

Object recognition is a crucial aspect of SOAR’s perception system, enabling it to identify and interact with various objects in its environment. To achieve robust and reliable perception, we combine 2D image-based object detection with 3D point cloud processing. We employed Ultralytics YOLO11[1] for real-time object detection, producing labeled bounding boxes for all recognized objects within the robot’s field of view. These bounding boxes are then used to segment and cluster point cloud data within the detected regions to compute 3D collision boxes and object poses, which are essential for downstream manipulation tasks. The pipeline is illustrated in Fig. 2.

¹ <https://www.ufactory.cc/xarm-collaborative-robot/>

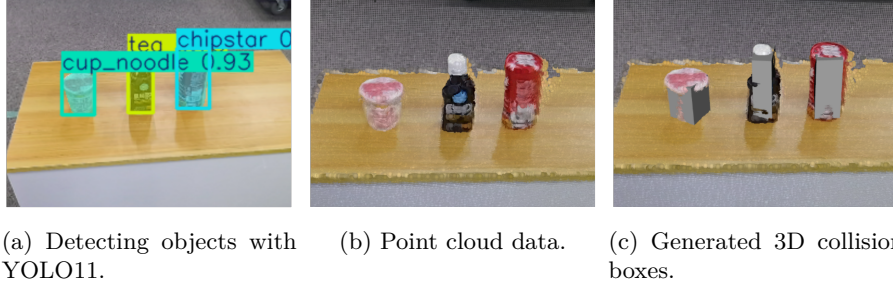


Figure 2: Object detection pipeline. YOLO11’s bounding box (Fig. 2a) and the point cloud data (Fig. 2b) are used to generate 3D collision boxes (Fig. 2c)

3.2 Speech Recognition And Interaction

Our previous system relied on EfficientWord-Net[2] for hotword detection, Google Dialogflow[3] for conversational understanding, and Google Text-To-Speech for vocal responses. Our current system mainly uses ChatGPT[4], known for its advanced natural language processing capabilities, to handle speech transcription, conversational understanding, and speech generation. The speech pipeline begins with Silero VAD [5] to segment incoming audio, followed by GPT-4o mini Transcribe [6] for speech-to-text conversion. The transcribed text is then processed by GPT-4.1 mini [4] dedicated to conversational understanding, which generates an appropriate response. The responses are converted back into audio using GPT-4o mini TTS.

In addition, we are exploring the use of the ChatGPT Realtime API[7], which supports low-latency, streaming interactions, and dynamic interaction where responses can be interrupted, updated, or redirected mid-generation based on new user input. To support this real-time conversational loop, we used Google’s WebRTC AEC3 [8] to suppress audio feedback from the robot’s own speakers, preventing them from being captured by the microphone. This enables continuous, overlapping dialogue, allowing the user to speak even while the robot is responding, resulting in more natural and fluid human-robot interaction.

3.3 Behavior

SOAR’s behavior system is built on the SMACH ROS package, which provides a hierarchical state machine framework forming the core of the robot’s decision-making architecture. Each SMACH state is designed as a modular, reusable component, allowing behaviors to be composed, extended, and reorganized for various tasks.

In standard task execution, state transitions follow predefined logic, but for the GPSR (General Purpose Service Robot) task, we use ChatGPT as a task planner to dynamically generate behaviors based on natural-language commands. Given a natural-language command, ChatGPT generates a sequence

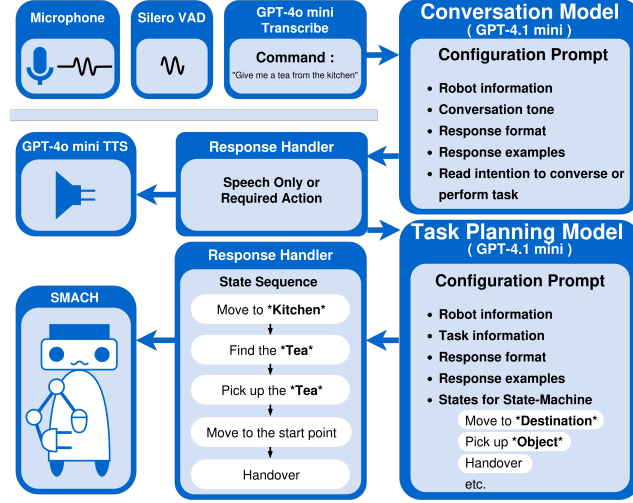


Figure 3: Behavior for GPSR task. When user says "bring me tea," the conversational model identifies it as an instruction, allowing the response handler to request plan from the task planning model. The model then returns a sequence of SMACH states for the response handler to execute.

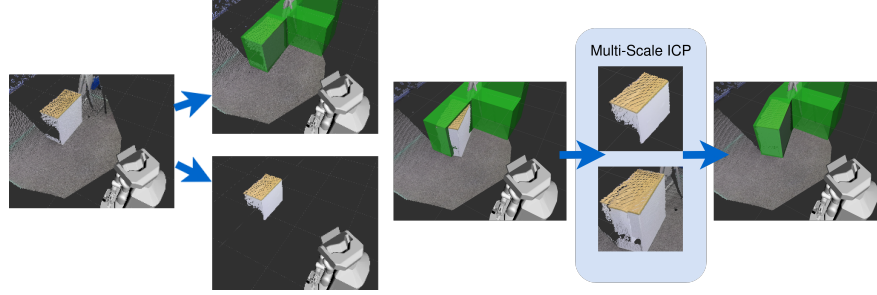
of states to execute, which are then executed in order without relying on the traditional state transition conditions. This approach enables flexible task planning and adaptation to open-ended commands. For more controlled execution, certain states themselves encapsulate internal state machines, allowing for structured sub-behaviors within a larger LLM-defined plan. As a result, some states function as primitive actions, while others manage complex behaviors, with the task planning model determining their order of execution while the robot's internal logic governs how each behavior is executed.

A key feature of our current architecture is the use of two separate GPT-4.1 mini models with distinct prompting strategies to clearly separate conversational interaction from task planning. All user input is first processed by the conversational model, which interprets intent, maintains dialogue context, and determines whether the user's request requires the robot to act. If an action is required, the conversational model passes the request to the task-planning model, which generates a sequence of SMACH states to execute the task.

The behavior pipeline along with its speech interaction for the GPSR task is illustrated in Fig. 3.

3.4 Scene fitting

In household environments, the scene around the robot is rarely static. While certain structures such as walls or large furniture remain fixed, many surfaces, such as tables and shelves where manipulation tasks occur, often shift slightly



(a) During setup, we create table's (b) When table pose shifts, the robot performs collision box and crop point cloud alignment between the reference model (top) as reference model. and the current environment (bottom) and updates the table's collision box pose.

Figure 4: Example of scene fitting a table.

due to human activities. These subtle changes can introduce significant risks during manipulation, as the mismatch between the robot's internal scene model and the actual environment can lead to collisions. To address this, we developed a scene fitting process that aligns its internal scene model with the current environment before performing manipulation tasks.

During the mapping phase, SOAR records the point cloud and the pose of key furniture surfaces involved in manipulation tasks, such as tables and shelves. At runtime, the robot captures the corresponding surface and applies a multi-scale ICP alignment to register the current observation with the previously stored model. The resulting transformation is used to update the robot internal scene model pose, ensuring that the motion planning accounts for slight environmental changes.

Fig. 4 illustrates an example of scene fitting for a table.

4 Contribution

Our recent contributions extend beyond the development of SOAR itself, focusing on fostering community growth and supporting research in robotics. We have focused on increasing our impact on the robotics community through open source software contribution and supporting research community initiatives. The following are some of our recent contributions:

- **Open-source software modules.**

We developed and released an open-source ROS1/ROS2 hardware interface² for VESC motor controller, which is also directly utilized in SOAR's mobile base system. For more of our open-source contributions, please refer to our GitHub repository³.

² <https://github.com/sbgisen/vesc>

³ <https://github.com/sbgisen>

- **ROS contributions.**

Members of our team actively contributed to official ROS repositories, including bug fixes, and feature enhancements.

- **Robotics conference and competition sponsorships.**

Our sponsorship includes domestic events such as the Robotics Society of Japan (RSJ) conference⁴, ROSCon JP⁵, and the Tsukuba Challenge⁶, as well as international engagement through the World Robot Summit⁷.

These activities reflect our ongoing commitment to advancing open, collaborative robotics research and support the global robotics community.

5 Conclusions and future work

In this paper, we presented SOAR, our evolving robot platform which will participate in the RoboCup@Home League. We detailed the hardware specification and introduced several software modules, such as object recognition and behavior system. Future work will extend and refine these modules, continuing to evolve SOAR to lay the technological and organizational foundation for future SoftBank robots. Each experiment and iteration enriches the collective understanding of robot cognition, perception, and interaction — the essence of our participation in RoboCup@Home.

References

1. Glenn Jocher, Ayush Chaurasia, and Jing Qiu. Ultralytics YOLO11, 2025.
2. R. Chidhambararajan, Aman Rangapur, S. Sibi Chakkaravarthy, Aswani Kumar Cherukuri, Meenalosini Vimal Cruz, and S. Sudhakar Ilango. EfficientWord-Net: An Open Source Hotword Detection Engine Based on Few-Shot Learning. *Journal of Information & Knowledge Management*, 21(04):2250059, 2022.
3. Navin Sabharwal and Amit Agrawal. Introduction to google dialogflow. In *Cognitive virtual assistants using google dialogflow: develop complex cognitive bots using the google dialogflow platform*, pages 13–54. Springer, 2020.
4. OpenAI. GPT-4 Technical Report, 2023.
5. Silero Team. Silero VAD: pre-trained enterprise-grade Voice Activity Detector (VAD), Number Detector and Language Classifier. <https://github.com/snakers4/silero-vad>, 2021.
6. OpenAI. Gpt-4o mini transcribe — models · openai platform. <https://platform.openai.com/docs/models/gpt-4o-mini-transcribe>, 2025. Accessed: 2025-11-06.
7. Introducing the Realtime API. <https://openai.com/index/introducing-the-realtime-api/>, 2024.
8. Google WebRTC team. Webrtc: Real-time communication project (chromium/external/webrtc). <https://chromium.googlesource.com/external/webrtc/>, 2025. Accessed: 2025-11-06.

⁴ <https://ac.rs-j-web.org/2023/>

⁵ <https://roscon.ros.org/jp/2025/>

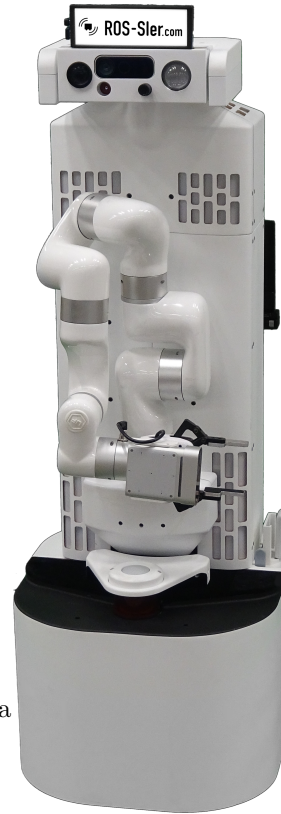
⁶ <https://tsukubachallenge.jp/202/organization/sponsors>

⁷ <https://wrs.nedo.go.jp/en/sponsor2020/>

SOAR Hardware Description

Specifications are as follows:

- Dimensions: (w) 500 mm (d) 530 mm (h) 1440 mm - 1790 mm
- Weight: 113 kg
- Max Speed: 1.2 m/s
- Battery: 25.6V 40Ah LiFePO4, 3 hours operation time
- Mobile Base: differential drive
 - Sensors:
 - * Hinson LE-50621 LiDAR
 - * LDROBOT LD06
 - * Wheeltec N100 9-axis IMU
- Torso unit: 1 DOF linear joint
 - PC: Intel NUC (NUC11PHKi7C)
- Manipulator: 7 DOF, 3.5 kg payload
 - Sensors:
 - * Azure Kinect DK RGB-D camera
- Head unit: 2 DOF pan-tilt
 - Sensors:
 - * Azure Kinect DK RGB-D camera
 - * NEEWER NW-410 condenser microphone
 - * GENIUSPY JM800S-30-360X zoom camera
 - * Seek Thermal CompactPRO Fast Frame
 - Display: THANKO C-79D21B
 - Speaker: YAMAHA VXS1MLB



SOAR Software Description

SOAR use the following open-source software components:

- Platform: Ubuntu 24.04 ROS2 Jazzy
- Navigation: Nav2, MPPI controller
- Manipulation: MoveIt2, MoveIt Task Constructor
- Object Recognition: YOLOv8
- Behavior Architecture: SMACH
- Speech Recognition: Silero VAD, Whisper
- Speech Generation: GPT-4o-mini-tts
- LLM: GPT-4o